

Microstructural heterogeneity and bonding strength of planar interface formed in additive manufacturing of Al–Mg–Si alloy based on friction and extrusion

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Abstract: Single-pass deposits of 6061 aluminum alloy with a single-layer thickness of 4 mm were fabricated by force-controlled friction- and extrusion-based additive manufacturing. The formation characteristics of the interface, which were achieved by using a featureless shoulder, were investigated and elucidated. The microstructure and bonding strength of the final build both with and without heat treatment were explored. A pronounced microstructural heterogeneity was observed throughout the thickness of the final build. Grains at the interface with Cu, {213}<111>, and Goss orientations prevailed, which were refined to approximately 4.0 μm . Nearly all of the hardening precipitates were dissolved, resulting in the bonding interface displaying the lowest hardness. The fresh layer, subjected to thermal processes and plastic deformation only once, was dominated by a strong recrystallization texture with a Cube orientation. The previous layer, subjected twice to thermal processes and plastic deformation, was governed by P- and Goss-related components. The ultimate tensile strength along the build direction in as-deposited and heat-treated states could reach 57.0% and 82.9% of the extruded 6061-T651 aluminum alloy.

Keywords: solid-state additive manufacturing; aluminum alloy; precipitate; crystallographic texture; heat treatment

1. Introduction

Metal-based additive manufacturing processes that use direct energy sources such as laser beams, wire and arc, and electron beams have already seen tremendous success. However, manufacturing aluminum alloys using conventional additive manufacturing with wire and arc, laser, or electron beams still presents numerous challenges [1–2]. Many solidification defects such as porosity and cracking cannot be eliminated entirely even if the innovative aluminum alloy powders with specific elements (such as Zr) or nanoparticles are developed based on some complex and unique technologies [3]. Friction-based additive manufacturing (FAM) processes derived from various friction welding principles are expected to address the aforementioned limitations due to the apparent advantages of solid-state bonding features for lightweight alloys [1].

To date, numerous FAM methods have been developed and investigated in the laboratory, and the approach developed by Aeroprobe Corporation, called “Additive Friction Stir Deposition (AFSD)”, is considered one of the most promising FAM methods [1,4]. The square bar of the starting material is inserted into a hollow, non-consumable tool at a high rotational speed and then pressed onto the surface of the substrate at a specific speed. It is essentially a friction surfacing process aided by a rotational shoulder, similar to the friction surface cladding process [5]. The concept of the

shoulder-assisted friction surfacing process is easy to understand, but innovative design ideas are required for the material feeding mechanism to realize the deposition process. Aeroprobe Corporation took the lead in developing its patented equipment first to verify the feasibility of this process [4] and designated the process as AFSD. Because this patented process is considered unique and AFSD may cause confusion with traditional friction stir additive manufacturing, the company named this process and their commercialized equipment MELD rather than AFSD [1]. Several scholars have conducted exploratory research on the deposition process with some metals using MELD equipment [1,6], demonstrating that it has enormous advantages in preparing large-scale high-performance components when compared to other existing FAM processes. The detailed technologies of patented MELD equipment, however, have not been made public.

It is well known that friction welding employs two types of control modes, force control and displacement control [7], which significantly affect the welding parameters and deposition processes, with force control generally considered more convenient than displacement control. The limited information published so far shows that the feed mechanism of the equipment MELD is displacement controlled. Therefore, we have independently developed a novel force-controlled spindle feed mechanism based on an authorized patent in China to realize the shoulder-assisted friction surfacing pro-

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cess. This process requires a hollow, rotatable shoulder containing a consumable feed rod (Fig. 1(a)). The feed rod is subjected to axial pressure while it rotates with the shoulder at the same speed and is pressed against the surface of the substrate by the hollow non-consumable shoulder. At the end of the consumable feedstock, the material rubs violently against the surface of the substrate and is softened by the resulting frictional heating. Under the influence of the extrusion of the precursor material, the softened metal continuously flows into the space between the shoulder and the substrate. Due to both the friction and the extrusion of the rotating shoulder, the surface layers of the substrate mix with the applied material and deform simultaneously, creating a strong and solid interfacial bond. The relative in-plane movement of the shoulder is used to achieve the desired layer thickness and shape, thus completing the deposition process (Fig. 1(b)). The concepts of "friction and extrusion" associated with the force control mode represent the critical features of this shoulder-assisted friction surfacing process. To highlight the special feature of this deposition process and to distinguish it from AFSD, because our processes are based on in-house developed equipment but not on the equipment from MELD, we refer to it here as friction extrusion additive manufacturing (FEAM).

MELD experiments for several metals, including nickel-based superalloys, aluminum, magnesium, and copper alloys, have previously been carried out [1,8–10]. Some fundamen-

al aspects of the thermal process, microstructure evolution, and properties of Al–Mg–Si alloys have been studied based on a tool with protrusions and a layer thickness of about 1 mm [10–16]. However, the research on the shoulder-assisted friction surfacing process by force control has not yet attracted enough attention. MELD and FEAM are different from fusion-based additive manufacturing processes because they are subject to frictional heating and severe plastic deformation. The thermal process and plastic deformation are inextricably linked to the material feeding mechanism (control mode), shoulder profile, starting materials (types, delivery conditions, and dimensions), and deposition parameters, which significantly affect the appearance of microstructure and properties of the final product. FEAM, in particular, is a newly developed and force-controlled friction-based additive process, with many critical mechanisms still unknown.

In the present study, a single-channel, double-layer Al–Mg–Si alloy was fabricated by FEAM using an independently developed force-controlled material feeding mechanism with a featureless shoulder. The mechanism of interface formation and the heterogeneity of deposition throughout the thickness of the final product, which was fabricated at a high deposition rate by increasing the thickness of the single-layer to 4 mm and the diameter of the starting material to 20 mm, were carefully investigated. The extent of restoration by post-deposition heat treatment on the microstructure and bonding strength was also evaluated.

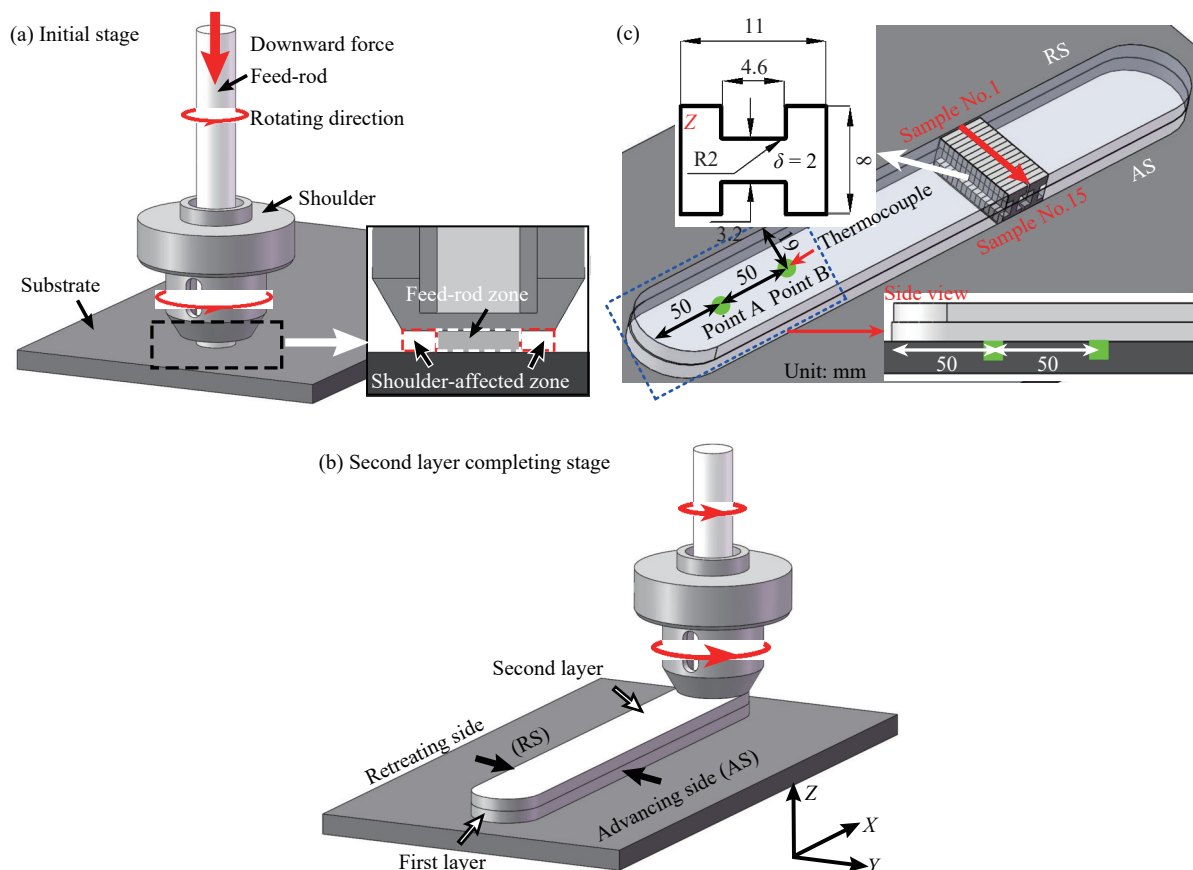


Fig. 1. (a, b) Schematic diagrams of the FEAM process; (c) dimensions of tensile specimens extracted along Z-direction (δ —Tensile sample thickness).

2. Experimental

The feed rods used in this study were commercial extruded rods of 6061-T651 aluminum alloy with a nominal diameter of 20 mm. The as-received composition of aluminum alloy 6061-T651 is shown in Table 1. The original temper condition of T651 indicates that it has been solution heat-treated, artificially aged, and stress-relieved by stretching. A 5 mm thick sheet of commercial Al 6061-T6 was used as the substrate.

Table 1. Chemical composition of starting material wt%

Mg	Si	Cu	Fe	Ti	Cr	Mn	Al
0.90	0.54	0.28	0.10	0.04	0.18	0.12	Bal.

All deposition experiments were carried out using a special solid-state friction additive manufacturing machine independently developed by Tianjin University. The main process parameters of FEAM include the rotation speed of the shoulder, the traverse speed, the thickness of the preplaced layer (the gap between the shoulder and the substrate or the previous layer), and the axial force acting on the feed rod of the consumable material. In this study, the deposition process was carried out with a rotation speed of 600 r/min and a traverse speed of 300 mm/min. The diameter of the featureless shoulder was 32 mm. The axial force applied to the consumable feed was 10 kN, and the preplaced thickness was 4 mm. The final build consisted of two layers with a length of 320 mm. The final build was also subjected to heat treatment (solution followed by tempering) to restore strength. After deposition, the final build was heat-treated (T6 tempering), dissolved at 525°C for 0.5 h in a muffle furnace, quenched in water, and immediately aged at 175°C for 8 h in a separate muffle furnace. Thermal cycling during deposition was measured using K-type thermocouples. Their fixed position is shown in Fig. 1(c). In this work, three directions, *X*, *Y*, and *Z*, were chosen to characterize the three-dimensional space of the final build; *X* is the direction of shoulder motion (the deposition direction or the length direction of the final build), *Y* is the width direction of the final build, and *Z* is the build direction (the thickness direction of the final build).

Microstructural features of the final build were quantitatively determined using electron backscattered diffraction (EBSD, JSM 7900 F- SEM with an EDAX-TSL system). Samples for EBSD measurements were prepared by electrolytic polishing with a 95 mL ethanol + 5 mL perchloric acid solution at 30 V for 12 s. To eliminate misorientation limits caused by orientation noise, a limit boundary misorientation cut-off of 2° was used in this study. Misorientation angles above 15° were defined as high-angle boundaries (HABs), while those between 2° and 15° were considered low-angle boundaries (LABs). Grain size measurements were obtained based on the EBSD maps by grain reconstruction without including the grains at the edges of the scan in the statistics. Here, kernel average misorientation (KAM) was introduced to estimate the homogeneity of the plastic deformation ex-

perienced by the final structure [17]. Geometrically necessary dislocations (GNDs) density was also introduced to quantify the change trends in dislocation density at different locations in the final build. The size and morphology of the nanoparticles were observed using a Tecnai G2F30 transmission electron microscope (TEM) at 200 keV. Energy-dispersive X-ray spectroscopy (EDX) was used to analyze the chemical composition of the particles. The exact position was determined by hardness determination.

Vickers microhardness tests along the centerline of each built layer, the interface, and the deposition centerline were performed using a digital display durometer (HVS-1000) with a load of 0.98 N for 15 s. The specimens for the tensile tests were taken along the *Z*-direction (the build direction of the final build), and their dimensions are shown in Fig. 1(c). The tensile tests in the *Z*-direction in the as-built and heat-treated conditions were performed using a micro tensile testing machine (INSTRON2710-004) with a constant crosshead speed of 0.2 mm/min. Fracture analysis was performed using a field emission scanning electron microscope (SEM, JEOL 7800).

3. Results

3.1. Macrostructure features

Fig. 2(a) depicts the final build's appearance as seen from the advancing side (AS) and retreating side (RS), respectively. Due to the friction and extrusion effect of the rotating featureless shoulder, fine arc-shaped corrugation was visible on the smooth surface, which was difficult to detect visually. The final build had no surface flaws. Each deposited layer had a consistent width of approximately 32 mm from start to finish, indicating that the FEAM process was stable and reliable. Fig. 2(b) depicts a macro view of the cross-section of the final build. Approximately 1–2 mm on either side of the bonding interface between the deposited layers could not be effectively bonded. The successfully bonded interface had a width of approximately 30 mm and was relatively compact; no internal defects (voids or cracks) were visible at this scale. In addition, the outer contours on both edges of the deposited layers were approximately semicircular. The contours on the RS had tiny saw teeth (see white arrows in Fig. 2(b)), while they were relatively smooth on the AS (see blue arrows in Fig. 2(b)). This is likely due to non-negligible differences in material flow on AS and RS during deposition.

Points c, d, e, and f in Fig. 2(b) are magnified to get a better look at the interface. The bonding interface presents a relatively flat and straight shape, indicating that the plasticized materials of the deposited layer did not mix with the previous layer during deposition. The interface zone should not be susceptible to etching, and it, therefore, appears bright white. No internal defects (holes or cracks) were observed at the interface in the center of the feed rod (Fig. 2(c)). However, holes were easily visible at the interface within the shoulder-affected zone at RS (Fig. 2(d) and (e)). In addition, a distinct dark gray line was discovered at the interface within the

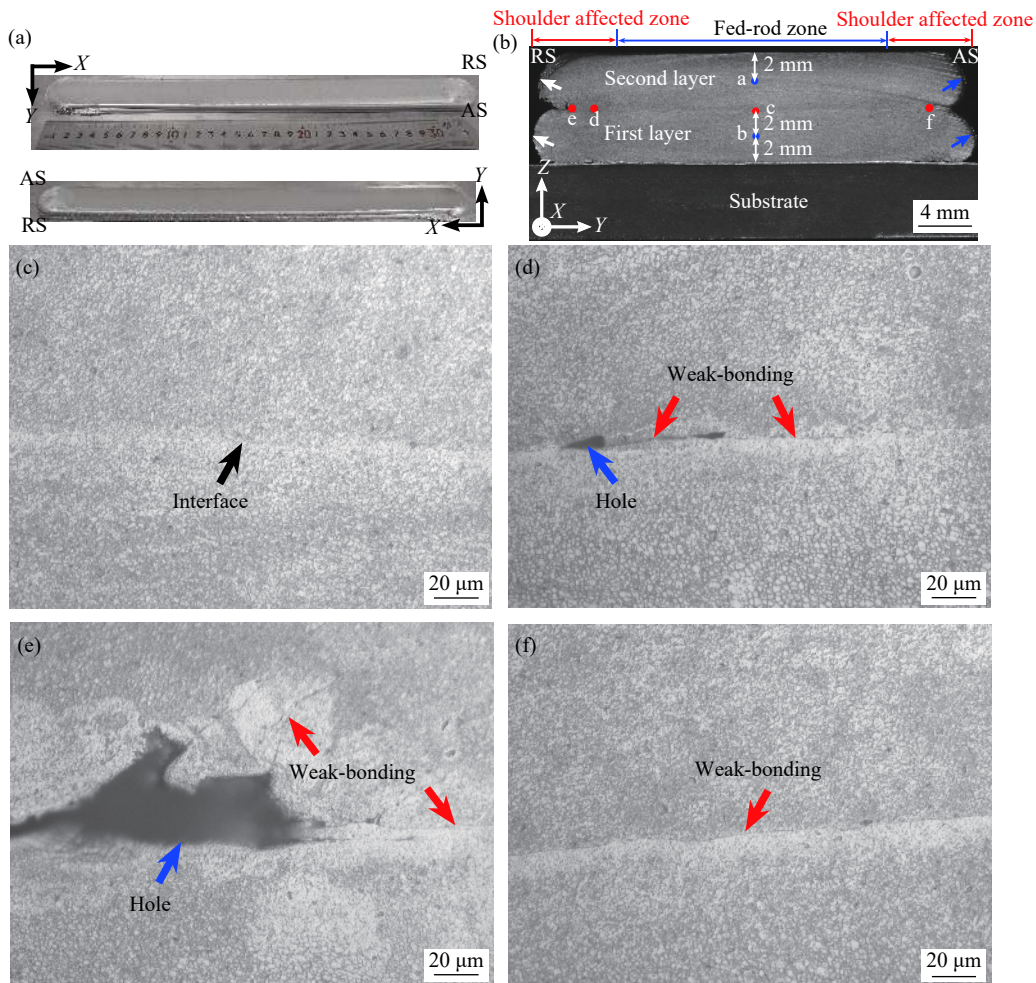


Fig. 2. (a) Appearance and (b) macro view cross-section of the final build; macro-view at various points of the interface in magnification: (c) point c, (d) point d, (e) point e, and (f) point f labeled in (b).

shoulder-affected zone, whether it was located at AS (Fig. 2(f)) or RS (Fig. 2(d) and (e)), while it was not found in the interface zone of the feed-rod zone. This gray line is supposed to be a weak-bonding defect, which will be examined in more detail below. Holes and weak-bonding defects that appeared at the interface within the shoulder-affected zone should be due to the relatively strong material flow and weak forging effect exerted by the shoulder.

3.2. Microstructure features

3.2.1. Starting material

Fig. 3(a) and (b) shows the maps of the inverse pole figure (IPF) and grain boundaries (GBs) of the as-received feed bar, where the direction of observation is along the extrusion direction (ED). The grains in the starting material developed with dominant $\langle 111 \rangle$ and $\langle 100 \rangle$ crystal orientations. The starting material exhibited a typical deformation structure with mean diameters of about $(22.9 \pm 11.5) \mu\text{m}$. A small amount of refined recrystallized grains were formed at the grain boundaries or triple junctions of the relatively large irregular grains. As shown in Fig. 3(b), HABs are represented by blue lines, while LABs are illustrated by red lines (similar to the following). The distribution of misorientation angles in the starting material was not random, as shown by the dashed

line in Fig. 3(c). A high fraction of LABs remained, accounting for about 57.5% of the total GBs. According to the inverse pole figures, as shown in Fig. 3(d), two fibers of $\langle 111 \rangle$ and $\langle 100 \rangle$ parallel to ED ([100]) were dominant in the starting material with intensities of 10.856 multiples of random distribution (MRD) and 4.903 MRD, respectively. In addition, $\langle 110 \rangle$ components parallel to the transverse direction (TD, [010]) were also formed with a weak intensity of 2.214 MRD.

Fig. 4 shows the corresponding textures as representative sections of the calculated orientation distribution functions (ODFs), superimposed on the main orientations of aluminum alloys after rolling and after recrystallization. The Miller indices and Euler angles of these orientations are listed in Table 2. It can be seen that the starting material was mainly dominated by four texture components with similar intensities: $\{213\}\langle 111 \rangle$, Cube ($\{001\}\langle 100 \rangle$), copper (Cu) ($\{112\}\langle 111 \rangle$), and P ($\{011\}\langle 566 \rangle$).

3.2.2. Final build in the as-deposited state

Fig. 5 depicts IPF maps for different areas of the final build in the as-deposited, with the direction of observation in the X-direction. These locations are labeled a, b, c, and d, respectively, in Fig. 2(b). The calculated grain size at the different locations is listed in Table 3. In the second layer, the

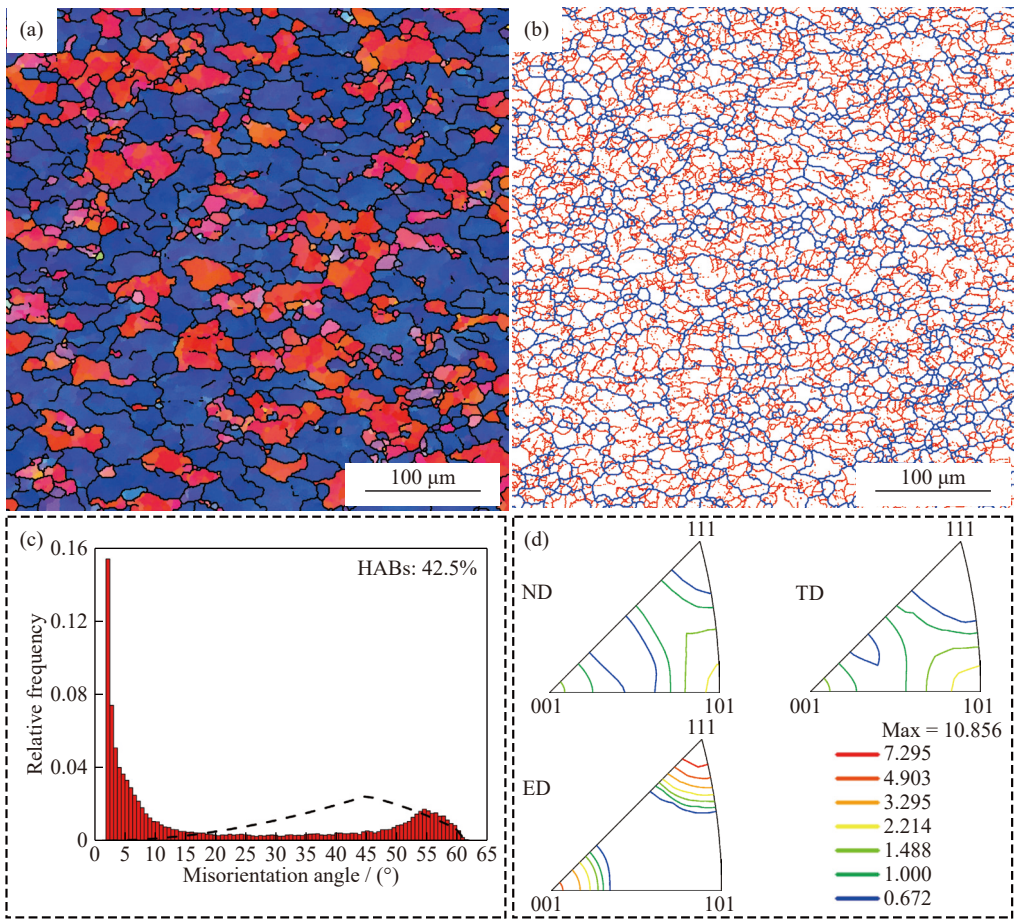


Fig. 3. (a) IPF map, (b) grain boundaries map, (c) misorientation-angle distribution (the dashed line represents a random distribution), and (d) inverse pole figures of the starting material. ND represents normal direction.

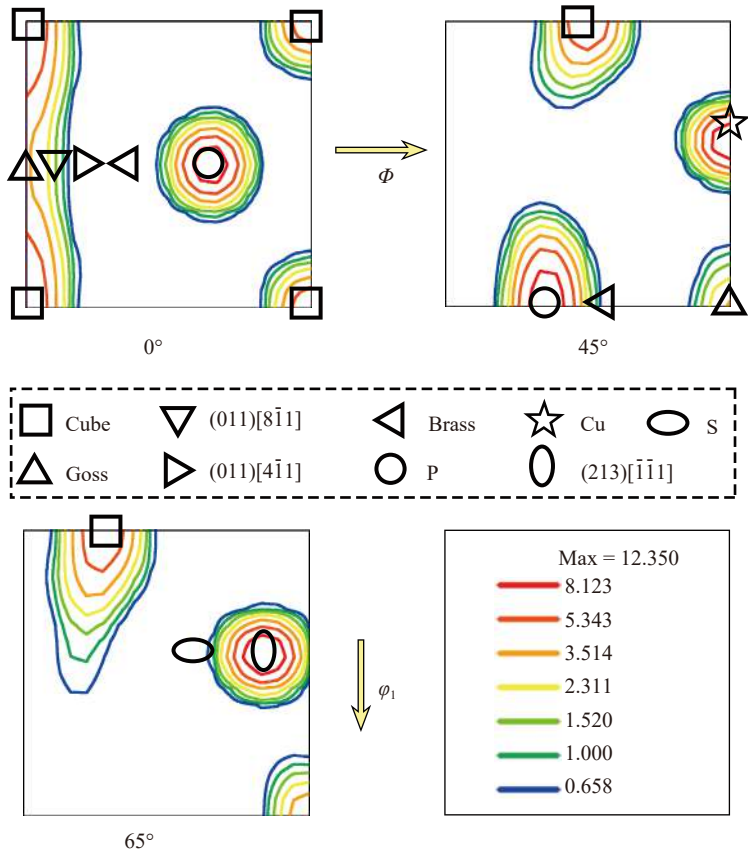


Fig. 4. ODFs superimposed on main orientations of Al alloys after rolling and after recrystallization of the starting material.

Table 2. Miller indices and Euler angles of main orientations of Al-alloys after rolling and after recrystallization [18–20]

Component	Miller indices, $\{hkl\}\langle uvw \rangle$	Euler angles / (°)		
		φ_1	Φ	φ_2
Cu	$\{112\}\langle 111 \rangle$	90	35	45
S	$\{123\}\langle 634 \rangle$	59	37	63
Brass	$\{110\}\langle 112 \rangle$	35	45	0
Goss	$\{110\}\langle 001 \rangle$	0	45	0
Cube	$\{001\}\langle 100 \rangle$	0	0	0
P	$\{011\}\langle 566 \rangle$	59	45	0
(011)[8 $\bar{1}$ 1]	$\{011\}\langle 811 \rangle$	10	45	0
(011)[4 $\bar{1}$ 1]	$\{011\}\langle 411 \rangle$	19	45	0
(213)[$\bar{1}$ 11]	$\{213\}\langle 111 \rangle$	75	37	63

grains formed in a nearly equiaxed shape with a diameter of about $(6.1 \pm 2.6) \mu\text{m}$ due to dynamic recrystallization during the deposition process (Fig. 5(a)). Some small grains in the initial recrystallization phase were also found at the boundaries or triple junctions of the grains with stable dimensions. The first layer contained many deformed and elongated grains, and the grain size $((7.5 \pm 3.5) \mu\text{m})$ was much larger. Surprisingly, grain refinement was more pronounced at the bonding interface of both the feed-rod and shoulder-affected zones. The difference between the two regions lies in the morphology and size of the grains. Fine recrystallized grains

Table 3. Calculated grain sizes of the final build in the as-deposited state

Zone	Grain size / μm
Starting material	22.9 ± 11.5
Second layer	6.1 ± 2.6
First layer	7.5 ± 3.5
Interface (feed-rod zone)	4.0 ± 1.9
Interface (shoulder-affected zone)	2.7 ± 1.1

are located along the grain boundaries or triple junctions of the relatively large, elongated deformed grains in the feed-rod zone, which have developed with dominant $\langle 111 \rangle$ and $\langle 100 \rangle$ crystal orientations. In contrast, the interface of the shoulder-affected zone exhibits finer equiaxed grains with grain refinement up to 67.5%.

Furthermore, the variation in grain misorientation angles should be considered. Fig. 6 depicts the GBs maps for various viewing positions in the final build. Fig. 7 shows the corresponding misorientation-angle distributions. Dynamic recrystallization was observed, along with a high fraction of HABs generated in the second layer, amounting to approximately 70.8% of the total GBs. Compared to the second layer, the fraction of HABs in the first layer decreased to 64.5%. Although there was noticeable grain refinement at the interface of the shoulder-affected and feed-rod zones, these two regions have nearly the same proportion of HABs as the

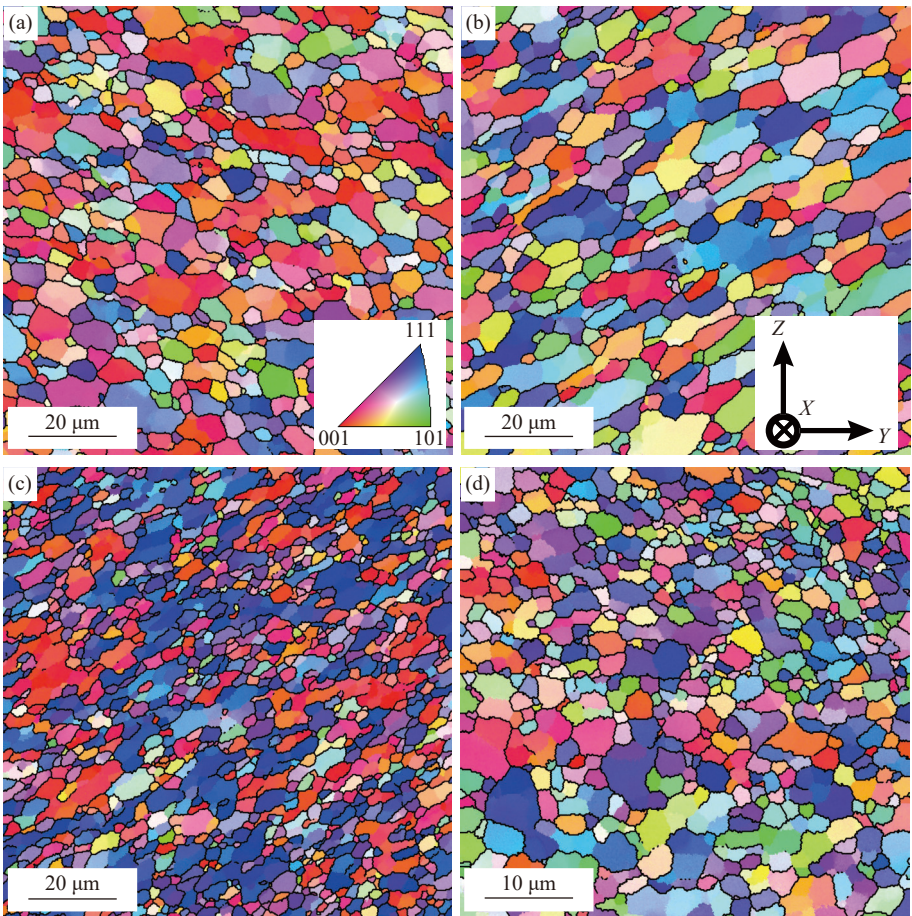


Fig. 5. IPF maps of the final build in the as-deposited state: (a) second layer, (b) first layer, (c) interface (feed-rod), and (d) interface (shoulder).

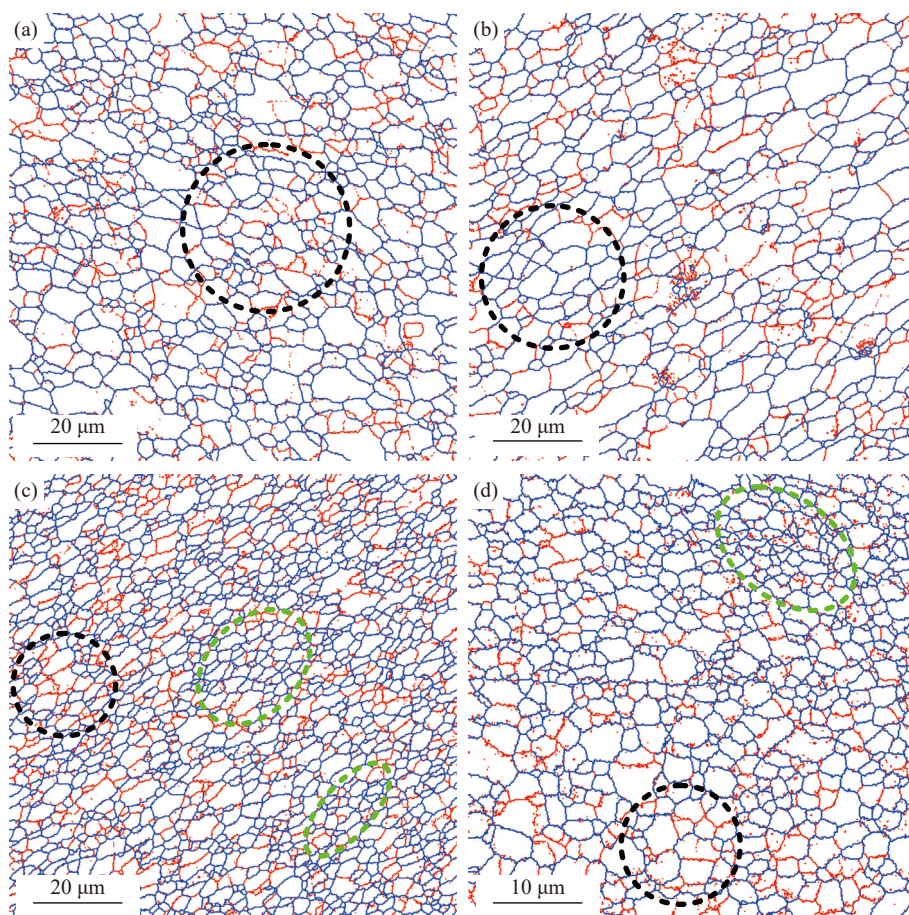


Fig. 6. GBs maps of the final build in the as-deposited state: (a) second layer, (b) first layer, (c) interface (feed-rod), and (d) interface (shoulder).

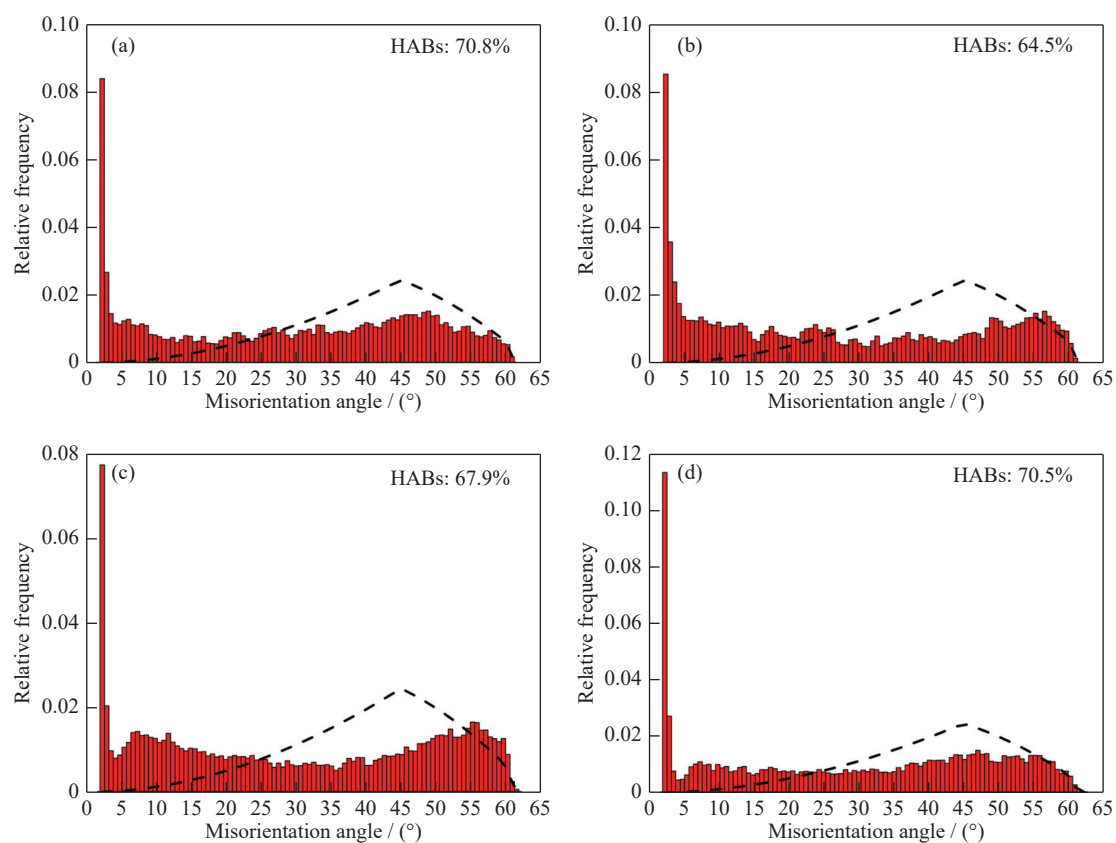


Fig. 7. Misorientation-angle distributions of the final build in the as-deposited state: (a) second layer, (b) first layer, (c) interface (feed-rod), and (d) interface (shoulder).

second layer. Furthermore, the orientation-angle distributions of all the four locations studied were not randomly distributed.

From the maps of KAM, the plastic deformation in the different regions of the final build exhibits apparent inhomogeneity (Fig. 8). The plastic deformation at the interface, whether in the shoulder-affected zone or feed-rod zone, is

more nonuniform than within the layer. As shown in Table 4, the GND density in the measured regions is about 1.6 to 3 times higher than that of the starting material due to the plastic deformation introduced during FEAM, and the GND density of the interface in the shoulder-affected zone is the largest and can reach 1.73 times that of the interface in the feed-rod zone.

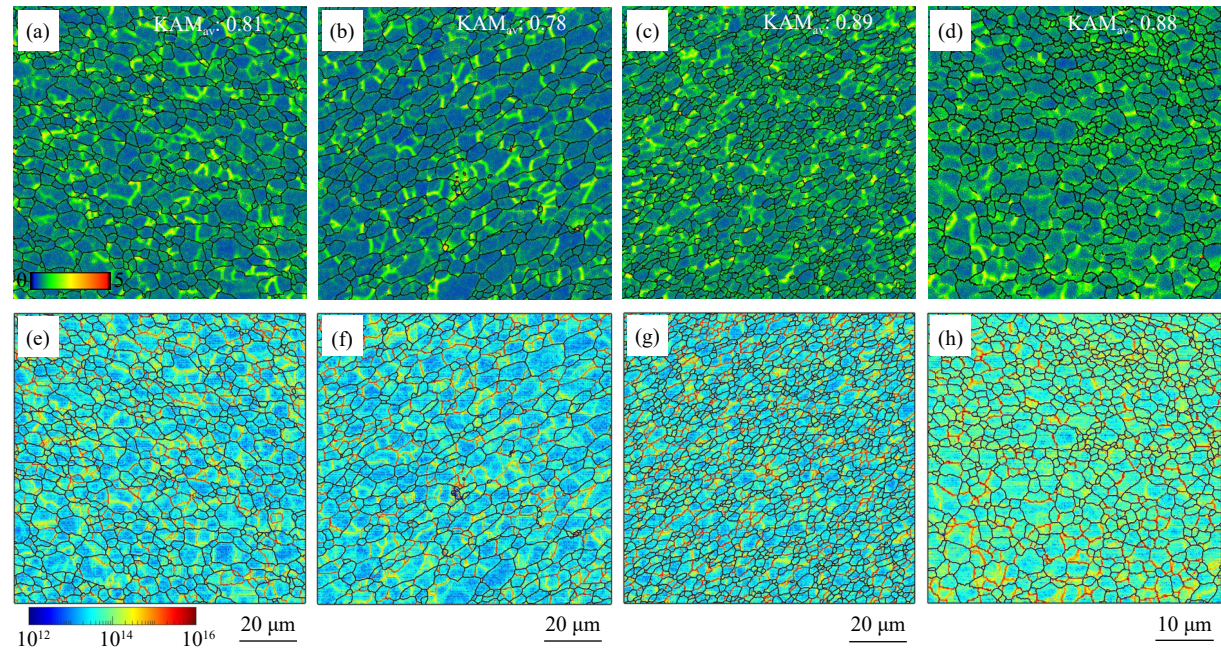


Fig. 8. (a–d) KAM and (e–h) GND maps of the final build in the as-deposited state: (a, e) second layer; (b, f) first layer; (c, g) interface (feed-rod); (d, h) interface (shoulder). KAM_{av} represents the average KAM.

The corresponding ODF in the final build was calculated and shown in Fig. 9. In addition to identifying and evaluating the position of the fiber types in Euler space, the volume fraction of the texture was also analyzed, as listed in Table 5. Compared to the starting material, there is a strong change in the texture components at the interface within the feed zone, which is dominated by Cu, Goss, $\{213\}\langle 111 \rangle$, and $\{011\}\langle 811 \rangle$ orientations. The peak intensity and volume fraction of Cu and $\{213\}\langle 111 \rangle$ orientations are almost the same and the largest, and Goss and $\{011\}\langle 811 \rangle$ orientations are next. It can be seen that a strong recrystallization texture of Cube orientation is formed in the second layer with a

volume fraction of 16.1%. In addition, S and $\{213\}\langle 111 \rangle$ components with volume fraction of 7.7% and 4.6%, respectively, are formed in this region. The major texture com-

Table 4. Calculated GND density of the final build in the as-deposited state

Observing zone	GND density / m^{-2}
Starting material	1.975×10^{13}
Second layer	3.180×10^{13}
First layer	3.177×10^{13}
Interface (feed-rod zone)	3.404×10^{13}
Interface (shoulder-affected zone)	5.894×10^{13}

Table 5. Volume fractions of orientations in starting material and final build within a cut-off angle of 15°

Component	Volume fraction / %				
	Starting material	Second layer	First layer	Interface (feed-rod)	Interface (shoulder)
Cu	11.7	1.6	0.7	19.8	2.3
S	0.2	7.7	1.9	4.0	12.0
Brass	0.1	1.2	0.3	0.9	5.5
Goss	5.3	0.5	4.4	13.8	0.9
Cube	11.0	16.1	0.0	0.1	2.0
P	19.1	2.2	12.9	2.1	4.7
$(011)[8\bar{1}1]$	0.8	1.2	21.5	8.0	3.8
$(011)[4\bar{1}1]$	0.0	1.4	4.3	1.3	4.6
$(213)[\bar{1}\bar{1}1]$	34.9	4.6	4.2	18.1	9.4

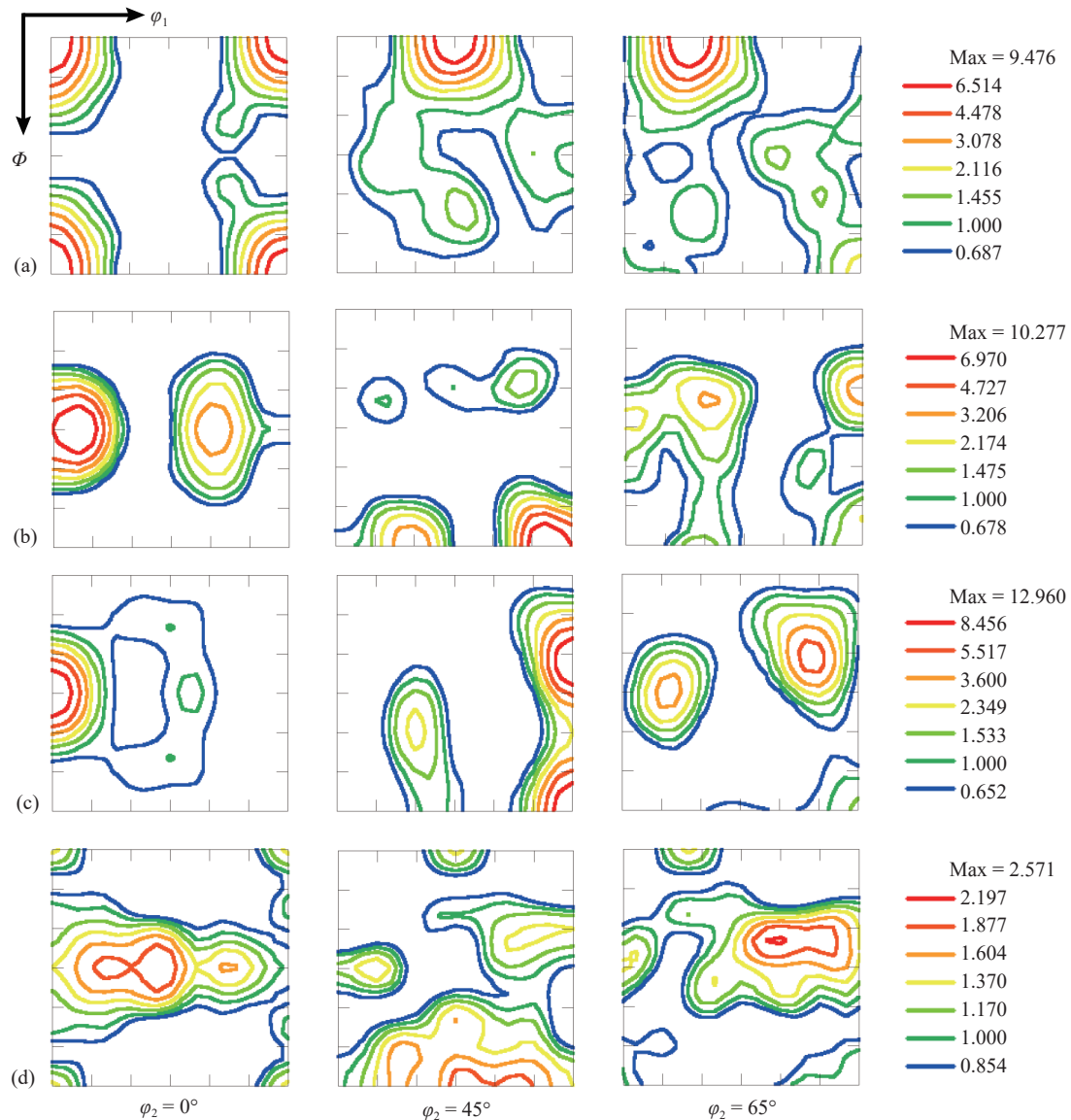


Fig. 9. ODFs of the final build in the as-deposited state: (a) second layer, (b) first layer, (c) interface (feed-rod), and (d) interface (shoulder).

ponents of the first layer are the $\{011\}\langle 811\rangle$ orientation and the P component with intensities of 10.277 MRD and 4.727 MRD, respectively. However, the total intensity of the textures at the interface within the shoulder-affected zone is reduced to 2.571 MRD. In this region, the S and $\{213\}\langle 111\rangle$ orientations prevail.

In addition to the grain structure and texture, the nano-scale particles also varied in the final build compared to the starting material. The evolution of the precipitates is closely related to the thermal cycles to which the deposited layers were subjected during FEAM. As shown in Fig. 10, the measured peak temperature at the interface between the first layer and the substrate within the feed-rod zone can reach 446–464°C during the deposition process of the first layer. In the deposition of the second layer, the maximum heating temperature at these test points can reach 405–415°C.

Fig. 11 depicts bright-field TEM images viewed along the $[001]$ direction of the Al matrix in the final build. The distribution of precipitates in the starting material showed a com-

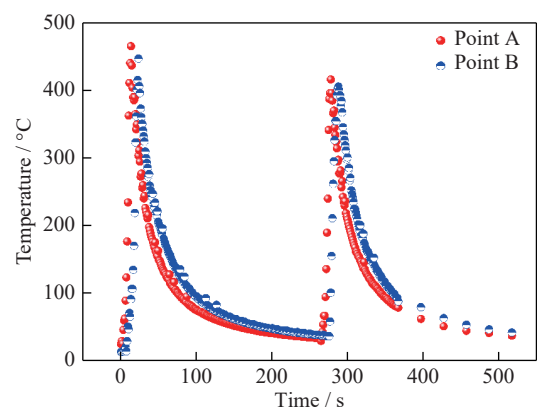


Fig. 10. Thermal cycling 50 mm (point A) and 100 mm (point B) from the start of the deposited layers.

paratively high density of needle-shaped precipitates (see black arrows) and a few rod-shaped precipitates (see white arrows) whose length was aligned along the $\langle 001\rangle$ Al direction (Fig. 11(a)). The needle-shaped precipitates (20–100 nm

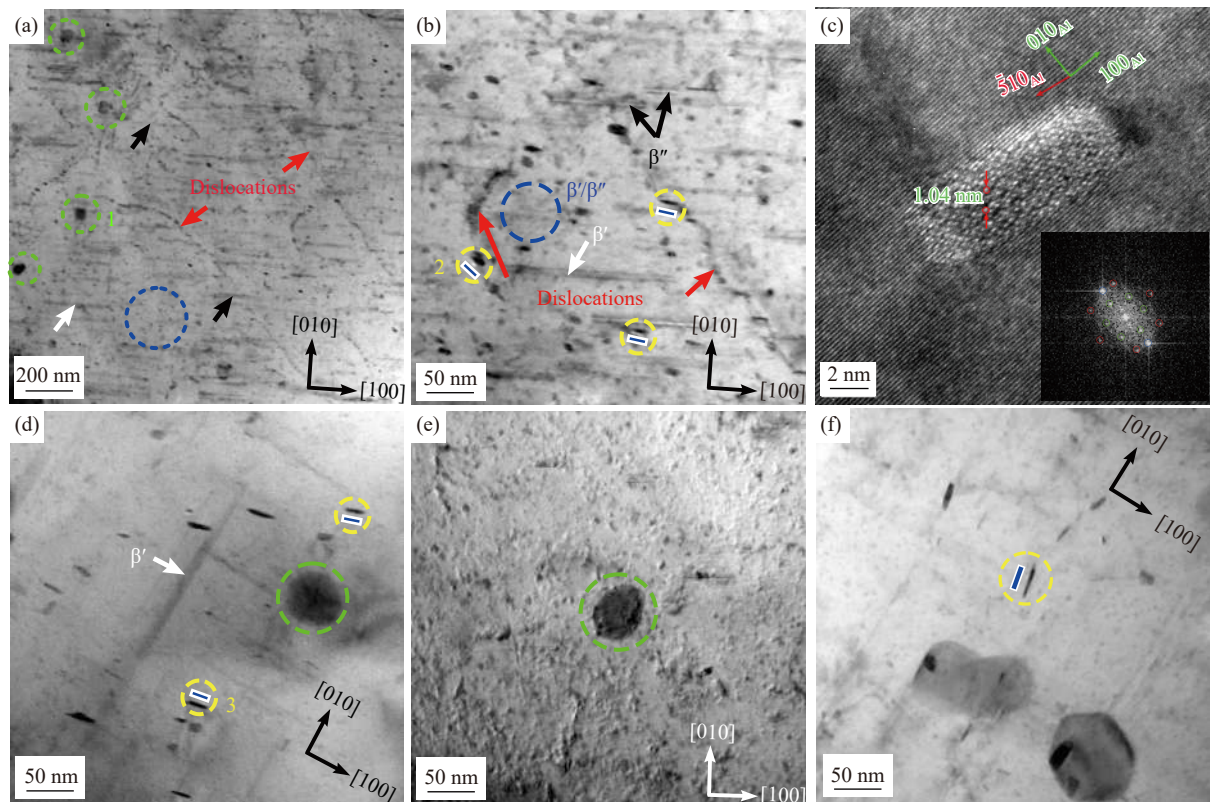


Fig. 11. (a, b) Bright-field TEM images of the as-received feed-rod; (c) high-resolution TEM image superimposed on the Fourier transform showing a precipitate with cross-section elongated along $\langle 510 \rangle_{\text{Al}}$; bright-field TEM images of final build in the as-deposited state: (d) second layer, (e) interface, and (f) first layer.

in length) are considered to be β'' , while the rod-shaped precipitates (50–300 nm in length) are considered to be β' , according to the typical features observed for both nanoscale precipitates [18–19]. In addition, string- and lath-shaped precipitates were observed along the dislocation lines (see red arrows and yellow circles in Fig. 11(a) and (b)). Their composition type is Al–Mg–Si–Cu, as identified by EDX measurements (Table 6). These precipitation types are similar to those reported in previous studies [21–23]. It can also be observed that the growth direction of these precipitates is not parallel to the $\langle 510 \rangle_{\text{Al}}$ direction (Fig. 11(c)). Therefore, based on the above interpretation and analysis, these lath-shaped copper-containing precipitates are considered to be Q' . The gray spots, as indicated by the dotted blue circles in Fig. 11(a) and (b) are considered to correspond to the cross-sectional shape of β'/Q' or β'' precipitates [24]. In addition, a small number of round-shaped precipitates with a diameter of approximately 50 nm are visible at low magnification (Fig. 11(a)). Their composition is Al–Fe–Mn–Si, as identified by EDX (Table 6). These iron-rich phases are thought to result from the breakup of massive intermetallics in the parent material during FEAM. However, these intermetallics have been shown not to reinforce the matrix [21–22].

In the second layer, almost no needle-shaped β'' was visible (Fig. 11(d)). It was replaced by a low density of thin lath-shaped Q' and coarsened rod-shaped β' , indicating that the maximum temperature within the layer reached the dissolution temperature of the needle-shaped precipitates. Some

Table 6. EDX analysis results of the points labeled in Fig. 11 at%

Point	Al	Mg	Si	Cu	Fe	Cr	Mn
1	89.91	1.52	1.21	—	2.72	1.73	2.82
2	89.93	6.19	3.39	0.47	—	—	—
3	71.27	16.48	9.95	2.18	—	—	—

iron-containing intermetallic phases were detected at the interface. However, nearly all the metastable or hardening precipitates were dissolved in this region (Fig. 11(e)) because the peak temperature experienced by the interface during deposition was above the dissolution temperature of β' . When depositing the second layer, the maximum temperature at the interface between the substrate and the first layer is 40 to 50°C lower than the peak temperature in the deposition process. Therefore, it can be seen that the undissolved precipitates in the first layer were further coarsened (Fig. 11(f)).

3.2.3. Final build in the heat-treated state

Fig. 12 depicts IPF maps of the final build in the heat-treated condition. T6 heat treatment resulted in recrystallization and grain growth in all observed regions. Significant grain growth occurred in the second layer and at the interface, both of which grew by about 2.5 μm compared to the as-deposited condition. In contrast, the first layer exhibited considerably less grain growth, increasing by approximately 1.3 μm . After the solid solution and aging treatments, the intermetallic phases were not noticeably altered, while the hardening precipitates were re-precipitated and dispersed in

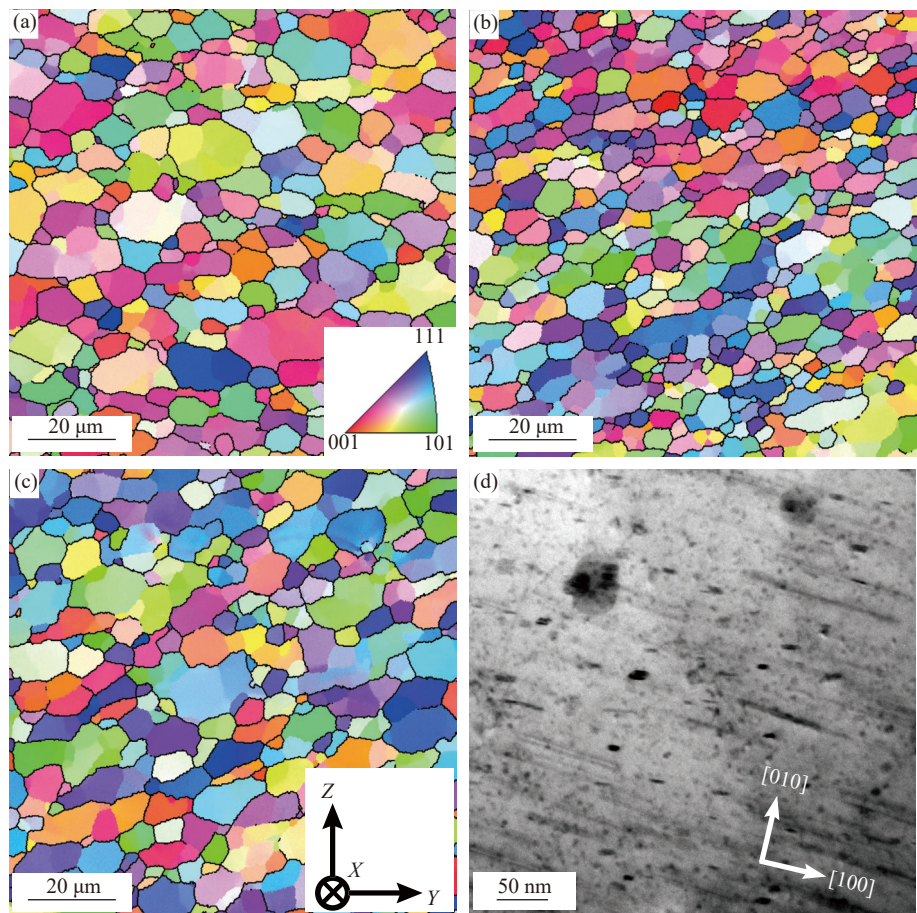


Fig. 12. (a–c) IPF maps and (d) bright-field TEM image of the final build in the heat-treated state: (a) second layer, (b, d) interface (feed-rod), and (c) first layer.

the matrix (Fig. 12(d)). However, string-type precipitates were no longer formed after T6 heat treatment compared to the starting material.

3.3. Mechanical performance

Fig. 13 depicts the Vickers hardness profiles along three horizontal lines on the cross-section of the final build in the as-deposited and heat-treated states. The distances between the measurement lines in the second and first layers and the upper surface were 2 and 6 mm, respectively. Compared with the as-received material (HV 113.1), the as-deposited speci-

men's hardness was remarkably reduced due to the dissolution of the β'' hardening precipitates. The microhardness profile was smoothly distributed in the second layer, with values ranging from HV 67.9 to 75.9, 60.0% to 67.1% of the hardness of the as-received feed rod. Compared to the second layer (HV 71.4 ± 1.9), the overall hardness of the first layer (HV 68.5 ± 3.1) was slightly lower, and the interface exhibited the lowest hardness. The hardness values of the interface ranged from HV 58.1 to 71.6, corresponding to 51.4% to 63.3% of the value of the as-received material. Furthermore, in the first layer and interface, the hardness within the feed

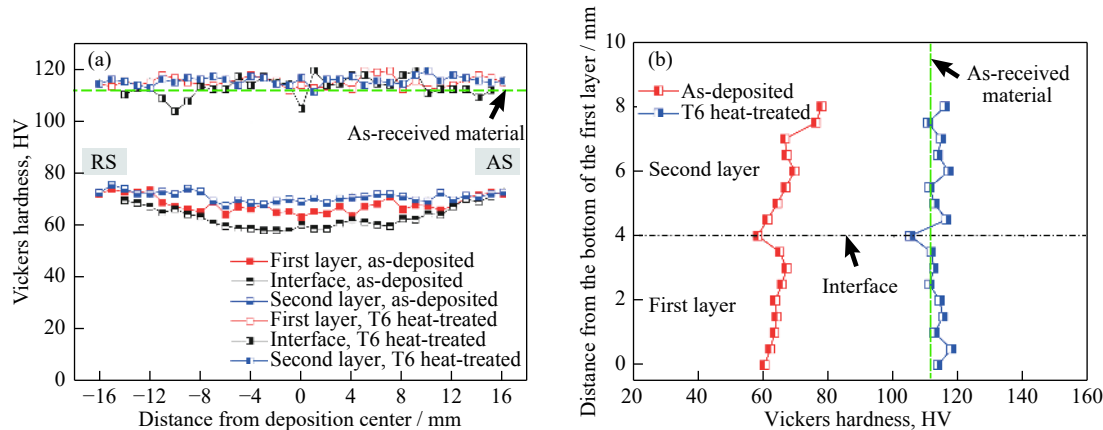


Fig. 13. Hardness profiles along the (a) horizontal and (b) build directions in the cross-section of the final build in the as-deposited and heat-treated states.

rod zone was lower than that in the shoulder-affected zone, and the latter was more pronounced. This indicates that the most intense temperature and plastic deformation occurred during the deposition process between the feed rod and the substrate or previously deposited layer.

After the T6 heat treatment, the re-precipitation of the hardening precipitates caused the hardness of the matrix to return to the level of the as-received material (Fig. 13(a)). The hardness of the first and second layers was not very large, with average values of HV (115.3 ± 2.1) and (115.8 ± 1.6), respectively. The hardness of the interface was slightly lower than that of the first and second layers but still comparable to the hardness of the material in the initial state.

Fig. 13(b) shows the hardness distribution along the thickness direction in the cross-section of the final build, with the measurement line coinciding with the centerline of the deposition. In the as-deposited state, the hardness gradually increased along the measurement line from the bottom of the first layer until it suddenly dropped to the lowest value of HV 58.1 near the interface. Upon entering the second layer, the hardness gradually increased again until it reached a maximum value of HV 77.7 at the surface of the second layer. After

heat treatment, the hardness was more uniform along the thickness direction and improved significantly. At almost all measured locations, the hardness was restored to higher than that of the starting material, except for the interface, which had the lowest hardness with a value of HV 105.

Fig. 14(a) and (b) depicts the tensile test results for the specimens sampled along the Z-direction in the as-deposited and heat-treated states. In the Z-direction, the width of a successfully bonded interface with the ultimate tensile strength (UTS) exceeding 175 MPa in both states is about 22 mm, as shown in Fig. 14(b). The UTS values of the as-deposited and heat-treated specimens were 57.0% and 82.9% (only the interface with valid bonding is counted) of the values of the as-received material (354 MPa), respectively. The specimens (labeled as No. 8) after tensile tests in the as-deposited and heat-treated states were observed, respectively, as shown in Fig. 14(a). The specimen without heat treatment fractured with a small deformation at the interface (see red rectangle in Fig. 14(a)). However, the heat-treated specimen broke from the interface with almost no deformation and exhibited a relatively straight fracture surface (see blue rectangle in Fig. 14(a)).

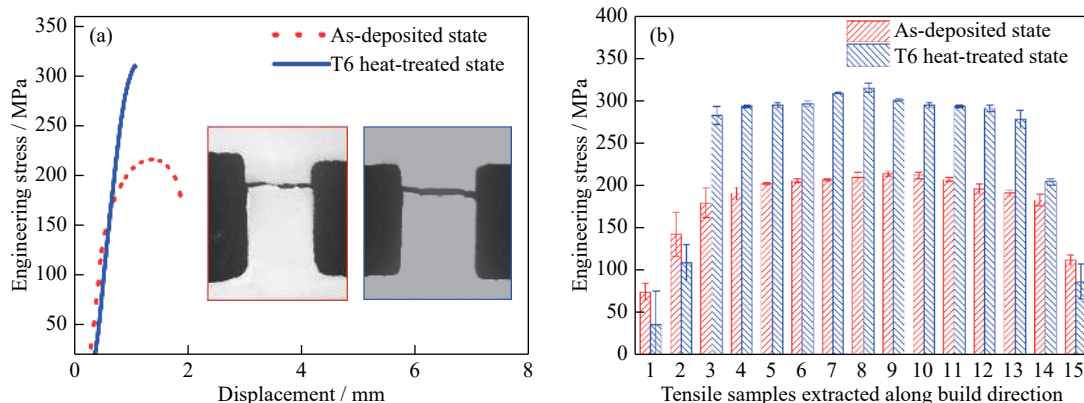


Fig. 14. Tensile test results of final build for specimens sampled along the Z-direction in the as-deposited and heat-treated states: (a) tensile stress–displacement curves; (b) UTS for specimens sampled along the Z-direction. Insets in (a) show the macro views of the samples (marked as No. 8) after tensile tests in the as-deposited (red rectangle) and heat-treated (blue rectangle) states.

The as-deposited specimen exhibited typical characteristics of quasi cleavage fracture (Fig. 15(a) and (b)), with the presence of both high-density tear ridge and a certain number of shallow dimples on the fracture surface (Fig. 15(c) and (d)). After heat treatment, the brittle fracture tendency of the specimen increased. The fracture morphology showed hybrid features with a high density of continuous small dimples and a thinner tear ridge, accompanied by a small number of large discontinuous dimples and secondary cracks (Fig. 15(f) and (g)).

4. Discussion

4.1. Interface microstructure effects on bonding strength

In the current study, FEAM was used to produce a nearly planar interface between the deposited layers, which is different from the non-planar interface produced by AFSD us-

ing a tool with protrusions [14]. Compared to the feed-rod zone, the as-deposited interface within the shoulder-affected zone displayed higher hardness but lower tensile strength along the Z-direction. This should be attributed to the weak-bonding defect formed at the interface during deposition. Regardless of the weak-bonding defect, the interface within these two regions exhibited significant differences in microstructural characteristics. An average 88.2% reduction in grain size was observed at the interface within the shoulder-affected zone. In contrast, this value was 82.5% for the feed-rod zone, suggesting a definite difference in the dynamic restoration process determined by the thermo-mechanical behavior experienced in these two regions. The dynamically recrystallized grain size (D_d) is considered as a function of the Zener–Hollomon parameter (Z_H), which depends upon temperature (T) and strain rate ($\dot{\epsilon}$), as described by the following formulas [25–26]:

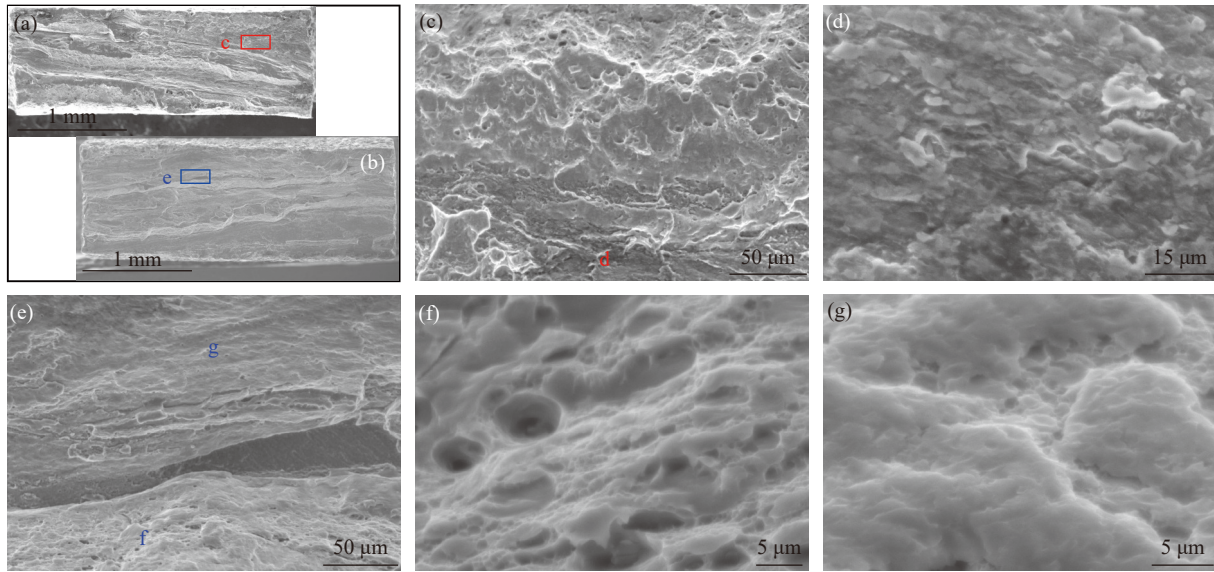


Fig. 15. (a, b) Macro and (c–g) micro views of fracture surfaces for specimens sampled along the Z-direction (marked as No. 8) in the (a, c, d) as-deposited and (b, e, f, g) heat-treated states.

$$D_d = cZ_H^{-n_d} \quad (1)$$

$$Z_H = \dot{\epsilon} \exp\left(\frac{Q_{\text{def}}}{RT}\right) \quad (2)$$

where c and n_d are constants, R is the gas constant, and Q_{def} is the activation energy for hot deformation. The temperature and strain rate experienced at the interface gradually moving away from the center of the feed rod decreased accordingly during FEAM. The interface within the shoulder-affected zone should be formed under a higher Z_H deformation condition than that within the feed-rod zone. It can be inferred that the recrystallization process of the latter was more sufficient under the condition of lower Z_H deformation. Accordingly, larger dynamically recrystallized grains were obtained, accompanied by a significantly lower dislocation density.

Such deformation conditions in these two regions are also considered to bring about differences in the restoration mechanism, although both displayed features of incomplete dynamic recrystallization. It can be observed that some fresh recrystallized grains almost entirely surrounded by HABs were generated as the pinch off of elongated deformed grains in most areas of the interface within the feed-rod zone (see green dashed ellipses in Fig. 6(c)), which presents a typical feature of geometric dynamic recrystallization (GDRX) [27]. In addition, a partially recrystallized necklace structure was found in the local region, consisting of a certain number of newly recrystallized grains formed by progressive lattice rotation (see white and black dashed circles in Figs. 5(c) and 6(c)) [15,27]. Progressive lattice rotation normally involves the progressive rotation of subgrains adjacent to pre-existing grain boundaries. This is the result of a gradient of misorientation that develops from the center to the edge of the parent grains as the applied strain increases [27]. It can also be observed that GDRX, which resulted in a large number of refined, initially recrystallized grains ($(2.3 \pm 0.8) \mu\text{m}$), occurred mainly at the bottom of the second layer above the bonding interface within the shoulder-affected zone (see

green dashed ellipses in Fig. 6(d)). The fraction of HABs increased to 77.6% in this region. Nevertheless, the top of the first layer below the bonding interface mainly underwent dynamic recovery accompanied by a considerable decrease in HABs (63.9%). There is also evidence of the continuous dynamic recrystallization (CDRX) process controlled by progressive lattice rotation (see white and black dashed circles in Figs. 5(c) and 6(c)) and the growth of recrystallized grains ($(3.1 \pm 1.3) \mu\text{m}$) in localized areas. A similar CDRX mechanism was also observed in the second layer and the first layer (see black dashed circles Fig. 6(a) and (b)). The original grain size (D_0) of the starting material and the deformation conditions are considered to determine the occurrence of GDRX. The critical strain (ϵ_{cr}) for GDRX is given by Eq. (3) [27],

$$\epsilon_{\text{cr}} = \ln\left(Z_H^{\frac{1}{m}} D_0\right) + K \quad (3)$$

where m and K are constants. When the starting material is deformed under a low Z_H condition or the original grain size is small, GDRX could occur at minor strains. Therefore, the heterogeneous thermo-mechanical coupling produced in the process of FEAM notably resulted in different restoration mechanisms.

Hardness and tensile tests exhibit that the interface where metallurgical bonding was achieved has the weakest mechanical properties of the final build, although it has the most refined grains and the highest dislocation density among the regions studied. This phenomenon shows that the effect of fine grain strengthening and dislocation strengthening at the interface is much less effective than the strength reduction due to the dissolution of hardening precipitates during FEAM. The strength along the Z-direction determined in this work is about 90% of the tensile strength of a friction stir welded joint of 6061-T6 made at a rotational speed of 1180 r/min and a welding speed of 95 mm/min [28]. The hardness in the middle of the second layer is about HV 5–8 higher than that of the first layer (Fig. 13(b)), which should be due to further precipitate dissolution or coarsening and slight grain

growth caused by the high temperature and plastic deformation to which the first layer was repeatedly subjected during the deposition of the second layer. The presence of a P component in this region could also support this (Fig. 9(b)), and a similar phenomenon was discovered in a previous study [29].

After T6 heat treatment, short-time solution treatment, and subsequent long-time aging, the tensile strength along the Z direction is greatly improved, but the brittle fracture tendency of the specimen is increased. The loss of strength after FEAM is inevitable for heat-treatable strengthened aluminum alloys, and post-deposition heat treatment processes applied on the final build should be necessary. However, abnormal grain growth must be avoided when restoring strength, which needs further research in the future.

4.2. Formation cause of weak-bonding in FEAM

In this study, the final build with a layer thickness of 4 mm was obtained with FEAM under an axial pressure of 10 kN. The width of the bonding interface produced was greater than the diameter of the feed rod and reached approximately the diameter of the assisted shoulder. However, the results of the tensile tests showed that the width for the metallurgical bonding was about 22 mm because of the weak-bonding defect that occurred during FEAM. FEAM differs significantly from friction surfacing in that the former requires the assistance of a rotating shoulder. Therefore, the preset layer thickness, the gap between the shoulder and the substrate or the previous deposition layer, is a critical factor in controlling the deposition process besides other process parameters such as axial pressure, rotation speed, and travel speed applied to the spindle. The frictional heat and plastic deformation generated between the feed rod and substrate or the previous deposit are considered to be the most significant heat generation sources in the FEAM process. Due to the heat generation and extrusion driven by the feed rod, the plasticized metal gradually flows into the shoulder-affected zone. As the thickness of the preset layer increases, it is not easy for the plasticized materials within the shoulder-affected zone to rotate together with the shoulder. The materials entering the shoulder-affected zone appear to be largely stationary and cannot generate heat through shear deformation. Deposition in this region can only be accomplished by the forging effect of the shoulder and heat generation through sliding friction at the interface between it and the substrate or the previous deposit. Thus, the shoulder-affected zone remains a deposition process dominated by compression, where the thermodynamic coupling is much lower than in the feed-rod zone. This can also be well verified from the differences in the microstructural features of the interfaces, in particular the grain orientation, between the two regions.

It can be observed that grains at the interface within the feed-rod zone with strong Cu, {213}<111>, and Goss deformation textures were predominant (Table 5). {213}<111> orientation has a relation with the brass component by a rotation of 38° around the [111] axis [30]. It is particularly noteworthy that grains of 6061 extruded rods used in this study are dominated by {213}<111> component. Thus, {213}

<111> orientation observed herein should be related to severe plastic deformation. At the interface within the shoulder-affected zone, the overall intensity of textures was remarkably decreased, with Cu, {213}<111>, and S orientations accounting for only 2.3%, 9.4%, and 12%, respectively, suggesting that this region has undergone less severe plastic deformation than the feed-rod zone.

In summary, the interface within the shoulder-affected zone is inferior to the feed-rod zone in terms of both heat generation and plastic deformation. The compression effect within the former region gradually decreases away from the rotating feed-rod. As a result, recrystallized grains that could eliminate the original contact area do not form when the new layer is deposited, and then weak-bonding defects are generated. However, the deformation and thermal processes experienced in the feed-rod zone and its adjacent location to the shoulder-affected zone are sufficient to achieve metallurgical bonding at the interface of these zones.

5. Conclusions

(1) The final build containing two layers of 6061 aluminum alloy with a single-layer thickness of ~4 mm was fabricated by a force-controlled FEAM process with a featureless shoulder, forming a planar interface that achieved a metallurgical bonding width of ~22 mm. The weak-bonding defect depicting a clear interfacial line was observed in the shoulder-affected zone because the high temperature and plastic deformation experienced were insufficient to generate recrystallized grains that could eliminate the initial interface.

(2) The interface with weak-bonding exhibited the most remarkable GND density and grain refinement (2.7 ± 1.1 μm) resulting from GDRX and progressive lattice rotation, which could reach 1.73 and 0.675 times of the interface without weak-bonding, respectively.

(3) The final build exhibited obvious heterogeneity of microstructure throughout the thickness. The interface had the largest reduction in hardening precipitates as it experienced the highest temperature and most severe plastic deformation. Strong deformation textures, such as Cu, {213}<111>, and Goss, were governed in the interface, while Cube orientation prevailed in the second layer and Goss related and P components were dominated in the first layer.

(4) The interface without weak-bonding consistently exhibited the lowest hardness both with and without heat treatment. The UTS along Z-direction in as-deposited and heat-treated states could reach 57.0% and 82.9% of the extruded 6061 aluminum alloy, respectively.

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2B03).

Conflict of Interest

The authors declare no conflicts of interest.

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